.things that Matter

Product Requirements Document (PRD)

Document Information	
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Company Name	**.things that Matter**
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Author	**Matome Mboweni, CTO**

1. Product Vision

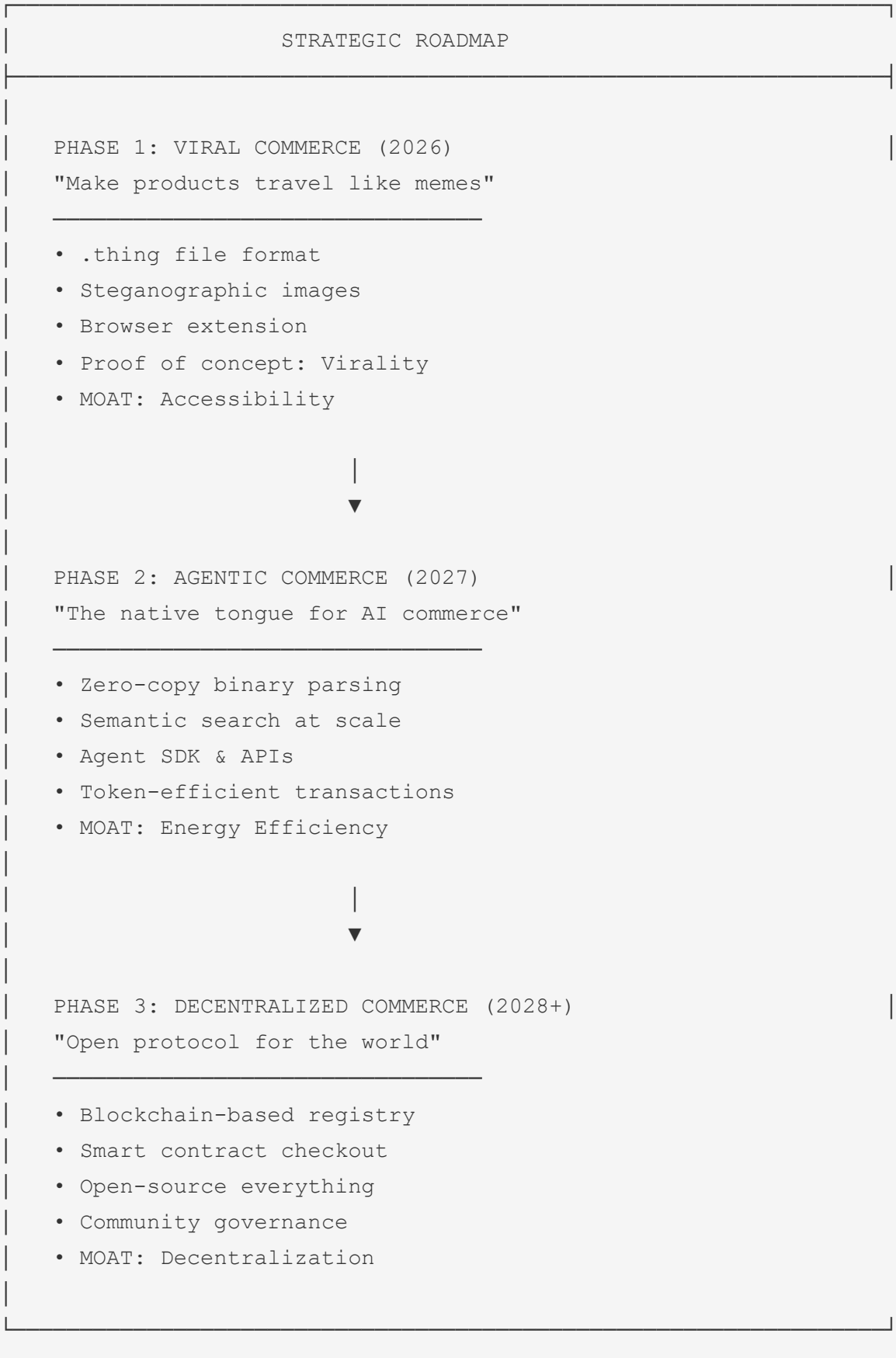
1.1 Vision Statement

Make commerce portable, viral, and universally accessible — for humans today, AI agents tomorrow, and a decentralized economy in the future.

1.2 Strategic Phases

Phase	Codename	Focus	Moat
Phase 1	**Viral Commerce**	Create .thing files and establish virality	**Accessibility**
Phase 2	**Agentic Commerce**	Optimize for AI efficiency	**Energy Efficiency**
Phase 3	**Decentralized Commerce**	Blockchain-based open infrastructure	**Decentralization**

1.3 Phase Overview


STRATEGIC ROADMAP
PHASE 1: VIRAL COMMERCE (2026)
"Make products travel like memes"
• .thing file format
• Steganographic images
• Browser extension
• Proof of concept: Virality
• MOAT: Accessibility
PHASE 2: AGENTIC COMMERCE (2027)
"The native tongue for AI commerce"
• Zero-copy binary parsing
• Semantic search at scale
• Agent SDK & APIs
• Token-efficient transactions
• MOAT: Energy Efficiency
PHASE 3: DECENTRALIZED COMMERCE (2028+)
"Open protocol for the world"
• Blockchain-based registry
• Smart contract checkout
• Open-source everything
• Community governance
• MOAT: Decentralization

2. Phase 1: Viral Commerce

2.1 Phase 1 Overview

Attribute	Value
Timeline	**2026 (11 months)**
Budget	**R72,000**
Goal	**Prove that products can travel virally across the internet**
Success Metric	**10,000 .thing files created, 100+ extension users**
Moat	**Accessibility — products available everywhere**

2.2 Phase 1 Problem Statement

Current State: - Products are trapped in platform silos - Sharing a product = sharing a link to a storefront - AI agents waste compute scraping HTML to understand products - No universal format for portable commerce

Desired State: - Products exist as self-contained digital objects - Sharing a product = sharing the product itself (in an image, link, or file) - Anyone can decode a product anywhere - Commerce becomes a layer of reality, not a destination

2.3 Phase 1 Requirements

2.3.1 Functional Requirements

ID	Requirement	Priority	Description
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P1-F01	.thing File Format	Must Have	Binary format using FlatBuffers schema
P1-F02	Commerce Compiler	Must Have	Transform product JSON → .thing files
P1-F03	CLIP Vectorization	Must Have	Generate 512-dim semantic embeddings
P1-F04	Ed25519 Signing	Must Have	Cryptographic tamper-proof seal
P1-F05	Zstd Compression	Must Have	Compress descriptions for portability
P1-F06	Polyglot Images	Must Have	Embed .thing in images (steganography)
P1-F07	XMP Metadata Beacon	Must Have	Fast discovery layer in image headers
P1-F08	matter:// URI	Must Have	Text-based distribution protocol
P1-F09	Shadow Index API	Must Have	Hash → artifact resolution service
P1-F10	Browser Extension	Must Have	Detect and display Matter products
P1-F11	Cloudflare R2 CDN	Must Have	Artifact storage and distribution
P1-F12	Batch Compilation	Should Have	Process 1000+ products at once
P1-F13	Merchant Dashboard	Could Have	Self-service product compilation

P1-F14	**Analytics**	**Could Have**	**Track .thing views and clicks**

2.3.2 Non-Functional Requirements

ID	Requirement	Target	Description
----	----------	--------	----------
P1-NF01	**Compile Latency**	**<500ms**	**Time to compile single product**
P1-NF02	**File Size**	**<50KB**	**Average .thing file size**
P1-NF03	**Extension Load**	**<100ms**	**Time to detect Matter content**
P1-NF04	**Signature Verify**	**<10ms**	**Client-side verification time**
P1-NF05	**CDN Availability**	**99.9%**	**Artifact availability SLA**
P1-NF06	**Steg Survival**	**≥50%**	**Platforms where payload survives**

2.3.3 Phase 1 User Stories

Merchant Stories:

ID	As a...	I want to...	So that...
----	---------	----------	----------
P1-US01	**Merchant**	**Upload my product catalog**	**My products become portable .thing files**

P1-US02	**Merchant**	**Download polyglot images**	**I can share shoppable images on social media**

P1-US03	**Merchant**	**Get matter:// links**	**I can share products in text-based channels**

P1-US04	**Merchant**	**See which products are compiled**	**I know my catalog is ready**

P1-US05	**Merchant**	**Update product prices**	**The .thing files reflect current pricing**

Consumer Stories:

ID	As a...	I want to...	So that...
----	---------	----------	----------
P1-US06	**Consumer**	**Install a browser extension**	**I can discover Matter products anywhere**
P1-US07	**Consumer**	**See product info on any image**	**I don't need to visit a store**
P1-US08	**Consumer**	**Click to buy directly**	**I can purchase without navigation**
P1-US09	**Consumer**	**Trust the price shown**	**I know it hasn't been tampered with**
P1-US10	**Consumer**	**Share a product with friends**	**They can also see the product info**

Developer Stories:

ID	As a...	I want to...	So that...
----	---------	----------	----------

P1-US11	**Developer**	**Access a compilation API**	**I can integrate Matter into my app**
P1-US12	**Developer**	**Read .thing files easily**	**I can build Matter-compatible tools**
P1-US13	**Developer**	**Verify signatures**	**I can trust the data integrity**

2.4 Phase 1 Acceptance Criteria

Milestone	Acceptance Criteria
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Phase 0 Complete	**Steganography tested on 20 platforms, ≥50% survival rate confirmed**
Compiler MVP	**Successfully compile JSON → .thing → polyglot image**
CLIP Working	**Semantic vectors embedded in .thing files**
Extension Alpha	**Detect XMP beacon, display product overlay**
75% Complete	**End-to-end demo: image → decode → product display**
Beta Launch	**5 merchants onboarded, 1000+ products compiled**
Public Launch	**Extension in Chrome Web Store, documentation complete**

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3. Phase 2: Agentic Commerce

3.1 Phase 2 Overview

Attribute	Value
Timeline	**2027**
Budget	**TBD (based on Phase 1 revenue)**
Goal	**Make .thing the most efficient format for AI agent commerce**
Success Metric	**100K+ agent transactions, documented efficiency gains**
Moat	**Energy Efficiency — cheaper for AI to use**

3.2 Phase 2 Problem Statement

Current State: - AI agents scrape HTML pages to understand products (expensive) - JSON APIs are verbose and require full parsing (inefficient) - Each platform has different data structures (fragmented) - Agents waste tokens on data extraction (costly)

Desired State: - .thing is the "native tongue" for AI commerce - Zero-copy field access means O(1) reads (fast) - Semantic vectors enable instant similarity search (smart) - Standardized format across all merchants (unified) - Agents use fewer tokens and less compute (cheap)

3.3 Phase 2 Requirements

3.3.1 Functional Requirements

ID	Requirement	Priority	Description
----	----------	----------	----------
P2-F01	Agent SDK	Must Have	Rust/JS libraries optimized for AI agents
P2-F02	Bulk Read API	Must Have	Efficient batch fetching of .thing files
P2-F03	Vector Search API	Must Have	Query products by semantic similarity
P2-F04	Streaming Compilation	Must Have	Real-time catalog sync for merchants
P2-F05	WASM Runtime	Must Have	Run compiler in any environment
P2-F06	Token Estimation	Should Have	Help agents budget API calls
P2-F07	Agent Analytics	Should Have	Track agent usage patterns
P2-F08	Rate Limiting	Should Have	Fair usage across agents
P2-F09	Caching Layer	Should Have	Reduce redundant fetches
P2-F10	GraphQL API	Could Have	Flexible querying for agents

3.3.2 Non-Functional Requirements

ID	Requirement	Target	Description
----	----------	--------	----------
P2-NF01	Read Latency	<5ms	Time to read single field from .thing

P2-NF02	**Vector Search**	**<50ms**	**Similarity search across 100K products**

P2-NF03	**Throughput**	**10K req/s**	**API requests per second**

P2-NF04	**Token Efficiency**	**90% reduction**	**vs. HTML scraping**

P2-NF05	**Compute Cost**	**80% reduction**	**vs. JSON parsing**

3.3.3 Phase 2 User Stories

AI Agent Stories:

ID	As an...	I want to...	So that...
----	----------	--------------	------------
P2-US01	**AI Agent**	**Read product prices in O(1) time**	**I minimize compute costs**
P2-US02	**AI Agent**	**Search products by semantic meaning**	**I find relevant items without keywords**
P2-US03	**AI Agent**	**Verify product authenticity**	**I don't transact with fake listings**
P2-US04	**AI Agent**	**Batch fetch product catalogs**	**I efficiently sync data**
P2-US05	**AI Agent**	**Estimate token usage before calls**	**I budget my operations**

Platform Stories:

ID	As a...	I want to...	So that...
----	---------	--------------	------------

P2-US06	AI Platform	Integrate Matter SDK	My agents can do commerce efficiently
P2-US07	AI Platform	Monitor agent transactions	I understand usage patterns
P2-US08	AI Platform	Set rate limits	I ensure fair resource usage

3.4 Phase 2 Efficiency Metrics

Comparison: Traditional vs. Matter Protocol

Operation	Traditional (HTML/JSON)	Matter (.thing)	Improvement
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Parse product page	~500ms, O(n)	~5ms, O(1)	**100x faster**
Read price field	Parse entire response	Direct byte offset	**Zero allocation**
Semantic search	External API call	Local vector math	**No API cost**
Verify authenticity	Trust platform	Cryptographic proof	**Trustless**
Token consumption	~2000 tokens/product	~50 tokens/product	**40x cheaper**
Memory usage	Full DOM/JSON	Memory-mapped file	**90% reduction**

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4. Phase 3: Decentralized Commerce

4.1 Phase 3 Overview

Attribute	Value
---	---
Timeline	**2028+**
Budget	**TBD (community-funded / token)**
Goal	**Open-source the protocol and decentralize the infrastructure**
Success Metric	**Protocol adoption by 1000+ independent nodes**
Moat	**Decentralization — no single point of control**

4.2 Phase 3 Problem Statement

Current State (Post Phase 2): - .things that Matter controls the signing keys - Shadow Index is a centralized service - CDN is our infrastructure - Users must trust our company

Desired State: - Anyone can become a signing authority - Product registry lives on blockchain - Artifacts stored on IPFS/Filecoin - Protocol governed by community - Trust is cryptographic, not institutional

4.3 Phase 3 Requirements

4.3.1 Functional Requirements

ID	Requirement	Priority	Description
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P3-F01	**Open Source Release**	**Must Have**	**All code MIT/Apache licensed**
P3-F02	**Decentralized Registry**	**Must Have**	**Blockchain-based product index**
P3-F03	**Multi-Signer Support**	**Must Have**	**Multiple trusted signing authorities**
P3-F04	**IPFS Integration**	**Must Have**	**Decentralized artifact storage**
P3-F05	**Smart Contract Checkout**	**Must Have**	**On-chain payment execution**
P3-F06	**Governance Token**	**Should Have**	**Community decision-making**
P3-F07	**Node Software**	**Should Have**	**Anyone can run a Matter node**
P3-F08	**Protocol Specification**	**Must Have**	**Formal spec for interoperability**
P3-F09	**Validator Network**	**Could Have**	**Decentralized compilation validation**
P3-F10	**Cross-Chain Bridge**	**Could Have**	**Multi-blockchain support**

4.3.2 Non-Functional Requirements

ID	Requirement	Target	Description
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P3-NF01	**Decentralization**	**No single point of failure**	**Full redundancy**
P3-NF02	**Censorship Resistance**	**Cannot be taken down**	**Distributed infrastructure**
P3-NF03	**Permissionless**	**Anyone can participate**	**Open protocol**

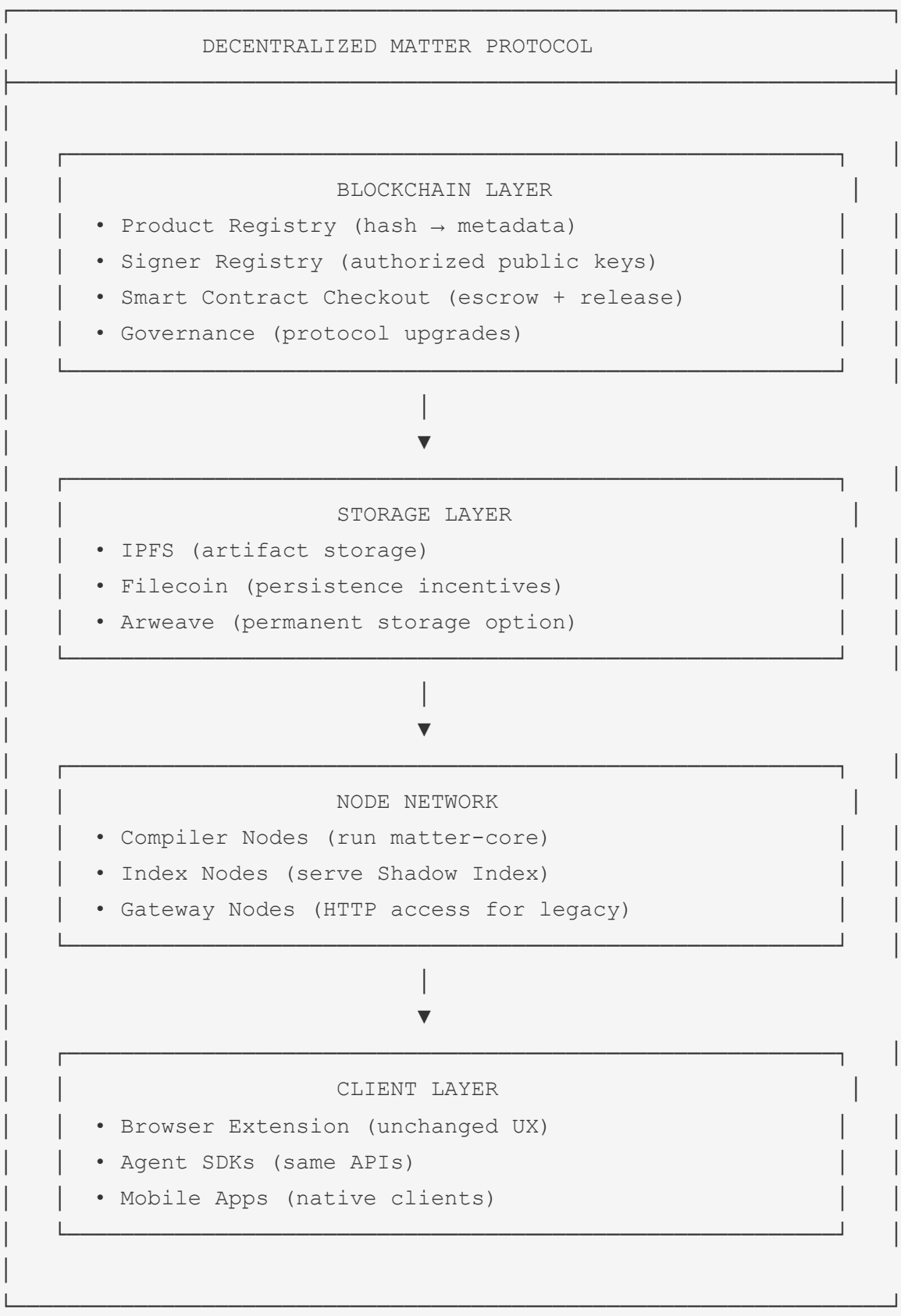
P3-NF04	**Backwards Compatible**	**Existing .thing files work**	**Smooth migration**

4.3.3 Phase 3 User Stories

Community Stories:

ID	As a...	I want to...	So that...
----	---------	----------	----------
P3-US01	**Developer**	**Fork and modify the protocol**	**I can innovate on commerce infrastructure**
P3-US02	**Node Operator**	**Run a Matter node**	**I earn fees and support the network**
P3-US03	**Merchant**	**Choose my signing authority**	**I'm not dependent on one company**
P3-US04	**Token Holder**	**Vote on protocol changes**	**I have a say in governance**
P3-US05	**Consumer**	**Trust math, not companies**	**My transactions are verifiable**

4.4 Phase 3 Architecture Vision


DECENTRALIZED MATTER PROTOCOL
BLOCKCHAIN LAYER
• Product Registry (hash → metadata)
• Signer Registry (authorized public keys)
• Smart Contract Checkout (escrow + release)
• Governance (protocol upgrades)
STORAGE LAYER
• IPFS (artifact storage)
• Filecoin (persistence incentives)
• Arweave (permanent storage option)
NODE NETWORK
• Compiler Nodes (run matter-core)
• Index Nodes (serve Shadow Index)
• Gateway Nodes (HTTP access for legacy)
CLIENT LAYER
• Browser Extension (unchanged UX)
• Agent SDKs (same APIs)
• Mobile Apps (native clients)

5. Feature Specifications (Phase 1 Detail)

5.1 .thing File Format

Purpose: Universal Commerce Container for portable product data.

Specification:

Field	Type	Required	Description
---	---	---	---
magic	uint32	Yes	0x54484E47 ("THNG")
version	uint8	Yes	Schema version (1)
id	string	Yes	Unique product identifier
hash	bytes	Yes	SHA-256 of canonical form
title	string	Yes	Product name
description	string	No	Product description (Zstd compressed)
price	uint64	Yes	Price in cents
currency	string	Yes	ISO 4217 currency code
sku	string	No	Merchant SKU
image_url	string	No	Primary image URL

clip_vector	float[512]	Yes	Semantic embedding
checkout_url	string	Yes	Purchase destination
merchant_id	string	No	Merchant identifier
created_at	uint64	Yes	Unix timestamp
expires_at	uint64	No	TTL for freshness
signature	bytes	Yes	Ed25519 signature
signer_pubkey	bytes	Yes	Signer's public key

File Size Targets:

Content	Size
---	---
Header + metadata	~500 bytes
CLIP vector	~2KB
Compressed description	~1-5KB
Signature	64 bytes
Total typical	**3-8KB**

5.2 Commerce Compiler

Purpose: Transform product data into .thing files and distribution assets.

Input:

```
json
{
  "id": "prod_123",
  "title": "Premium Wireless Headphones",
  "description": "High-fidelity audio with 30-hour battery...",
  "price": 14999,
  "currency": "USD",
  "image_url": "https://example.com/headphones.jpg",
  "checkout"url": "https://store.example.com/checkout/prod123"
}
```

Output:

```
json
{
  "hash": "QmYwAPJzv5CZsnA625s3Xf2nemtYgPpHdWEz79ojWnPbdG",
  "thing_url": "https://r2.thingsthatmatter.com/QmYwAP...thing",
  "image_url": "https://r2.thingsthatmatter.com/QmYwAP...jpg",
  "matter_uri": "matter://QmYwAPJzv5CZsnA625s3Xf2nemtYgPpHdWEz79ojWnPbdG",
  "gist": {
    "title": "Premium Wireless Headphones",
    "price": 14999,
    "currency": "USD"
  }
}
```

Compilation Pipeline:

1. **Validate** — Check input against schema 2. **Fetch Image** — Download product image 3. **Vectorize** — Send to CLIP API, receive 512-dim vector 4. **Compress** — Zstd compress description 5. **Serialize** — Build FlatBuffers binary 6. **Hash** — SHA-256 of canonical binary 7. **Sign** — Ed25519 signature 8. **Embed** — Steganographic image creation 9. **Upload** — Push to Cloudflare R2 10. **Index** — Record in Shadow Index

5.3 Browser Extension (Lens)

Purpose: Detect and display Matter products on any webpage.

Features:

Feature	Description
XMP Detection	Scan images for Matter metadata beacons
URI Interception	Handle matter:// protocol links
Overlay Display	Show product card on hover/click
Signature Verification	Validate Ed25519 in WASM
Local Cache	IndexedDB storage for resolved products
Settings	Enable/disable, visual preferences

UI Components:



Permissions Required:

Permission	Reason
activeTab	Read page content for detection
storage	Cache resolved products
webRequest	Intercept matter:// URIs

5.4 Shadow Index API

Purpose: Resolve product hashes to artifact locations.

Endpoints:

Method	Endpoint	Description

| --- | --- | --- |
POST	**/v1/compile**	**Compile product JSON to .thing**
POST	**/v1/batch**	**Compile multiple products**
GET	**/v1/resolve/{hash}**	**Get artifact URLs for hash**
GET	**/v1/gist/{hash}**	**Get minimal product data**
GET	**/v1/health**	**Service health check**

Example: Resolve

```
http
GET /v1/resolve/QmYwAPJzv5CZsnA625s3Xf2nemtYgPpHdWEz79ojWnPbdG

Response:
{
  "hash": "QmYwAPJzv5CZsnA625s3Xf2nemtYgPpHdWEz79ojWnPbdG",
  "thing_url": "https://r2.thingsthatmatter.com/QmYwAP.thing",
  "image_url": "https://r2.thingsthatmatter.com/QmYwAP.jpg",
  "gist": {
    "title": "Premium Wireless Headphones",
    "price": 14999,
    "currency": "USD"
  },
  "created_at": 1706000000,
  "expires_at": 1737600000
}
```

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6. Success Metrics

6.1 Phase 1 KPIs

Metric	Target	Measurement
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.thing files created	10,000	Database count
Unique products compiled	5,000	Distinct SKUs
Extension installs	100+	Chrome Web Store
Active extension users	50+	Weekly active
Merchants onboarded	5-10	Signed agreements
Steganography survival	≥50%	Platform testing
Compile success rate	>99%	API metrics
Average compile time	<500ms	API metrics
Extension detection rate	>95%	Test suite

6.2 Phase 2 KPIs (Future)

Metric	Target	Measurement
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Agent SDK downloads	**1,000+**	**Package registry**
Agent transactions	**100K+**	**API logs**
Token efficiency gain	**40x**	**Benchmark tests**
Compute cost reduction	**80%**	**Benchmark tests**
Vector search queries	**1M+**	**API metrics**

6.3 Phase 3 KPIs (Future)

Metric	Target	Measurement
---	---	---
Independent nodes	**1,000+**	**Network discovery**
Open source contributors	**100+**	**GitHub metrics**
Protocol forks	**50+**	**GitHub forks**
Decentralized transactions	**1M+**	**Blockchain data**

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7. Dependencies & Integrations

7.1 External Dependencies

Dependency	Purpose	Phase	Risk
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Replicate API	CLIP inference	P1	Medium
Cloudflare R2	Artifact storage	P1	Low
Supabase	Database	P1	Low
Vercel	API hosting	P1	Low
Chrome Web Store	Extension distribution	P1	Low
IPFS	Decentralized storage	P3	Medium
Ethereum/Polygon	Smart contracts	P3	Medium

7.2 Integration Points

Integration	Type	Phase
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Merchant product catalogs	Inbound data	P1
Social media platforms	Image distribution	P1
AI agent frameworks	SDK integration	P2

Blockchain networks	**Protocol layer**	**P3**

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8. Constraints & Assumptions

8.1 Constraints

Constraint	Impact
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R72,000 Year 1 budget	**Lean development, free tiers**
Solo developer	**Sequential work**
11-month Phase 1 timeline	**Extended milestones**
No security audit budget	**Self-review initially**

8.2 Assumptions

Assumption	Validation
---	---
Steganography survives ≥50% of platforms	**Phase 0 testing**
CLIP API remains affordable	**Monitor pricing**

Chrome allows extension	**Review policies**
Merchants want portable products	**Beta feedback**
AI agents will adopt efficient formats	**Market trends**

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9. Glossary

Term	Definition
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.thing file	**Universal Commerce Container binary format**
Matter Protocol	**The overall system for portable commerce**
Polyglot Image	**Image file that is also a valid .thing file**
Shadow Index	**Hash-to-artifact resolution service**
Lens	**Browser extension for Matter detection**
CLIP Vector	**512-dimension semantic embedding**
Zero-Copy Access	**Reading data without memory allocation**
Steganography	**Hiding data within images**

XMP Beacon	**Metadata header for fast discovery**

matter:// URI	**Custom protocol for text-based sharing**

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10. Document Approval

Role	Name	Signature	Date
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CEO	**Sobuhle Nyambali**		**/_/2026**
CTO	**Matome Mboweni**		**/_/2026**

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